

Week 1 - Assignment
Data Preprocessing & Exploratory Data Analysis
(EDA)

Titanic Dataset

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1. Introduction

Machine learning models rely heavily on high-quality data. Raw datasets often contain missing values, duplicate records, inconsistent formats, and outliers that can reduce model performance. Therefore, data preprocessing and exploratory data analysis (EDA) are essential steps before developing predictive models.

The objective of this assignment is to preprocess the Titanic dataset, explore its characteristics using statistical analysis and visualizations, and prepare it for future machine learning model development. The target variable for this dataset is **Survived**, which indicates whether a passenger survived the Titanic disaster.

2. Dataset Overview

The Titanic dataset contains information about passengers who traveled aboard the RMS Titanic. The dataset includes demographic information, ticket details, travel class, family relationships, and survival status.

Dataset Information:

- Number of Rows: **891**
- Number of Columns: **12**
- Target Variable: **Survived**
- Numerical Features: PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
- Categorical Features: Name, Sex, Ticket, Cabin, Embarked

The dataset was initially inspected using the `head()`, `info()`, `describe()`, `isnull()`, and `uplicated()` functions to understand its structure and identify potential data quality issues before preprocessing.

```
df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  -
 0   PassengerId   891 non-null    int64
 1   Survived      891 non-null    int64
 2   Pclass        891 non-null    int64
 3   Name          891 non-null    object
 4   Sex           891 non-null    object
 5   Age           714 non-null    float64
 6   SibSp         891 non-null    int64
 7   Parch         891 non-null    int64
 8   Ticket        891 non-null    object
 9   Fare          891 non-null    float64
10   Cabin         204 non-null    object
11   Embarked      889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

3. Initial Data Inspection

The initial inspection revealed that the dataset contained **891 rows and 12 columns**. Missing values were identified in the **Age**, **Cabin**, and **Embarked** columns. Specifically, the Age column contained **177 missing values**, Cabin contained **687 missing values**, and Embarked contained **2 missing values**.

No duplicate records were found in the dataset. Data types were also verified to distinguish between numerical and categorical variables before applying appropriate preprocessing techniques.

4. Data Preprocessing

Is section me screenshots lagana:

- Missing Values
- Duplicate Check
- Boxplots
- Encoded Dataset
- Scaled Dataset

Task	Method Used	Reason
Age	Median	Robust to outliers
Cabin	Dropped	77% missing values
Embarked	Mode	Only 2 missing values
Duplicate Records	None Found	No action required
Outliers	Retained	Genuine passenger information
Sex	Label Encoding	Binary category
Embarked	One-Hot Encoding	Multiple categories
Scaling	StandardScaler	Normalize numerical features

Assignment Questions

Q: Why Median for Age?

Because Age contains outliers, the median provides a more robust estimate than the mean.

Q: Why was Cabin removed?

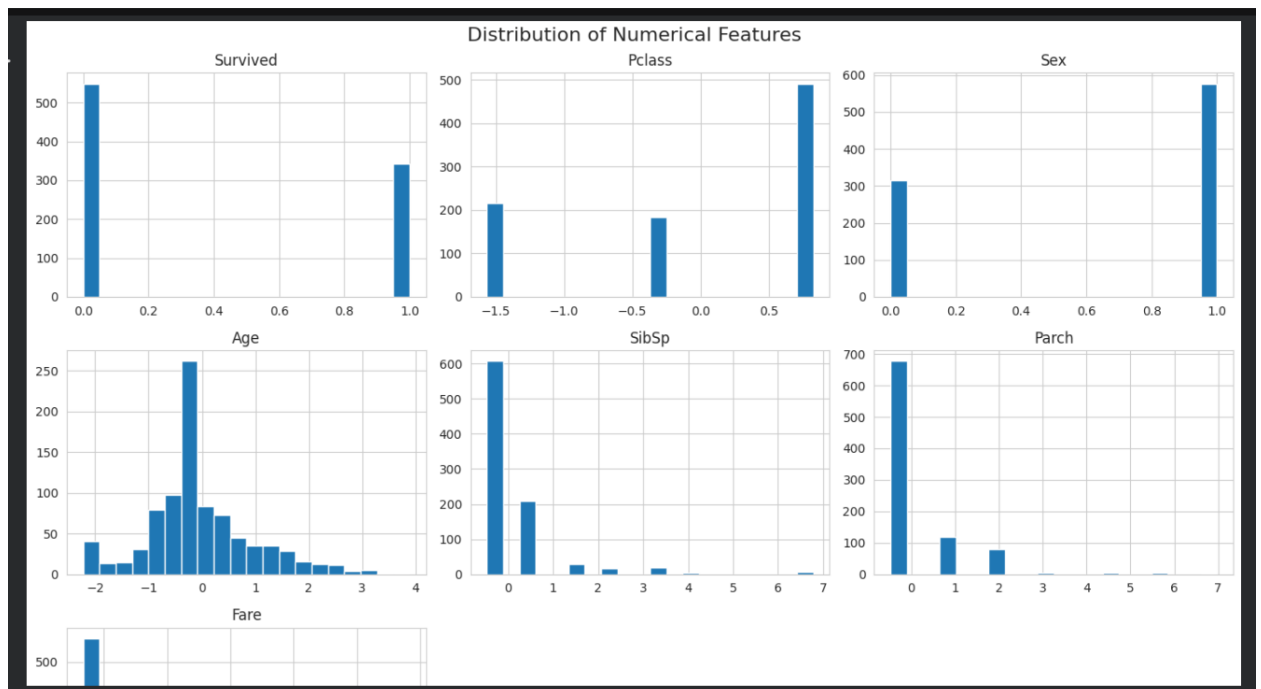
Because approximately 77% of its values were missing, making reliable imputation impractical.

Q: Why were outliers retained?

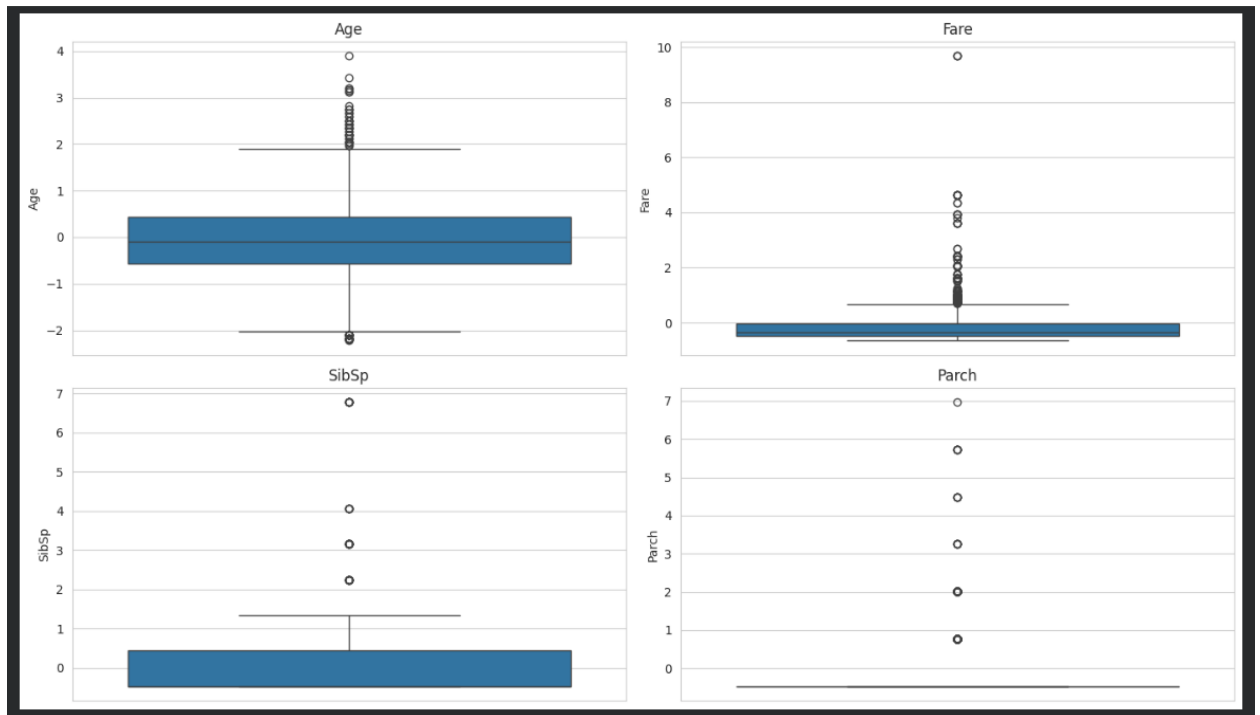
The detected outliers represented genuine passenger information, such as high ticket fares and large family groups, rather than data entry errors.

5. Exploratory Data Analysis (EDA)

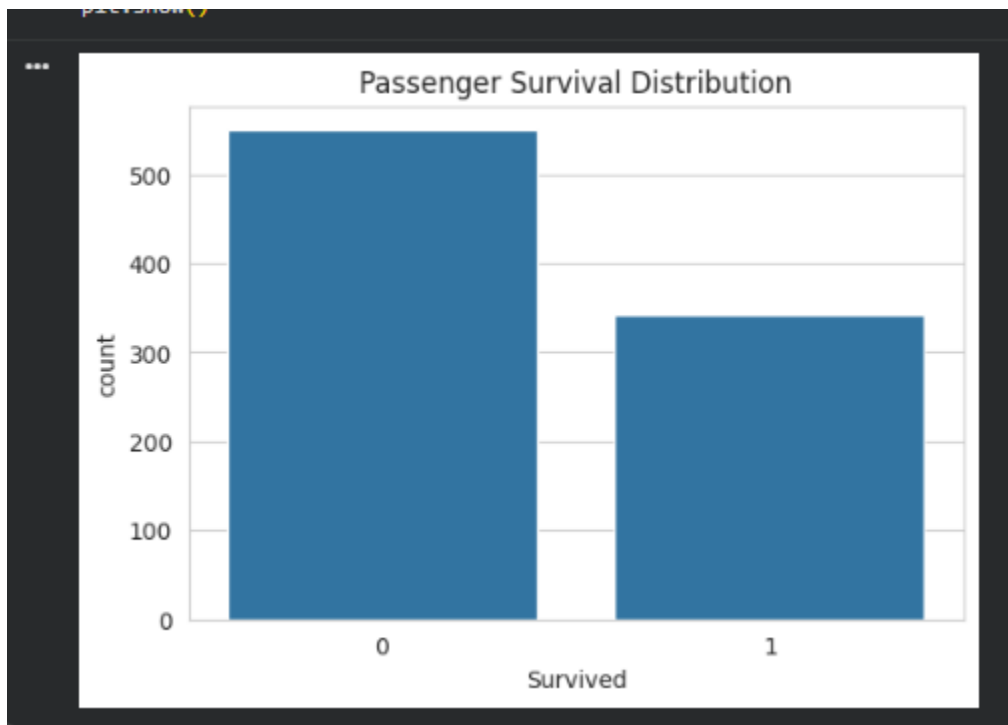
- Histogram



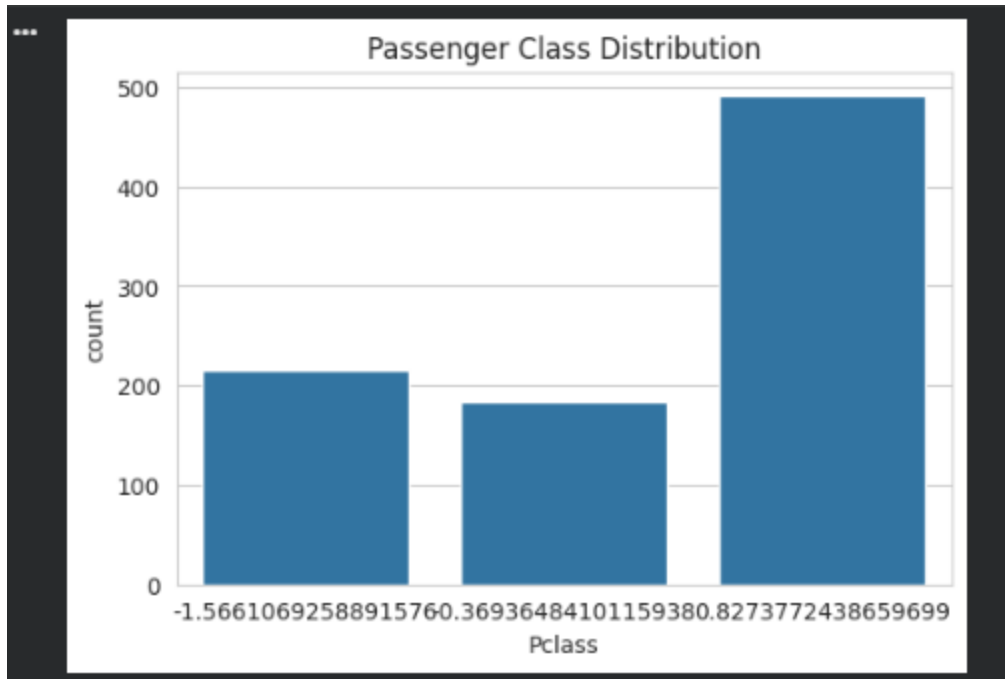
- Boxplot



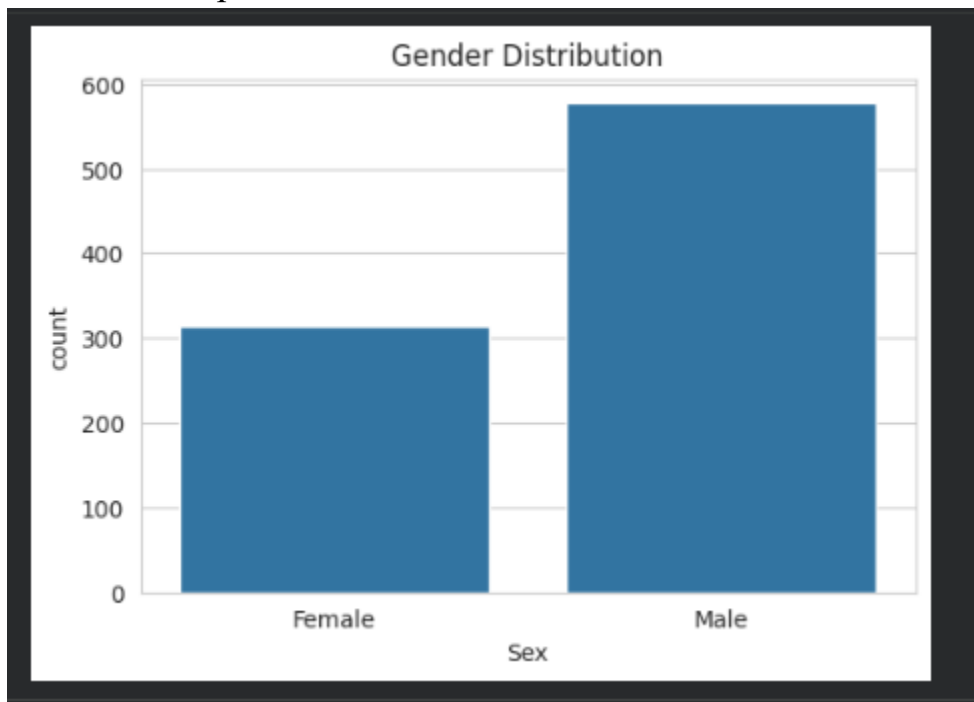
- Survival Countplot



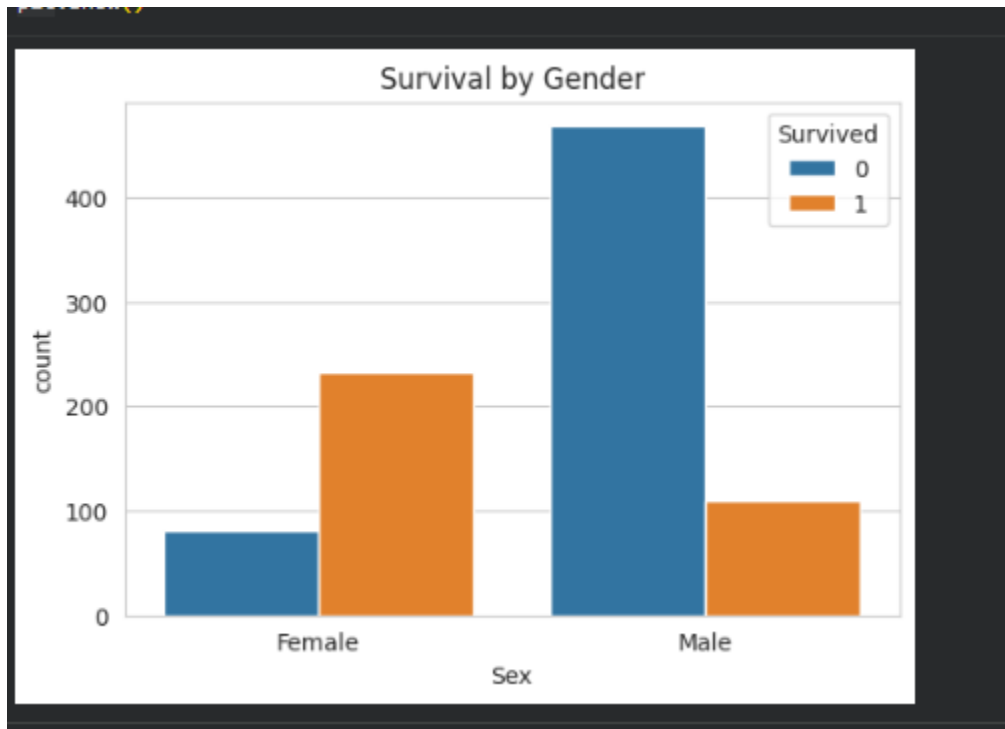
- Passenger Class Countplot



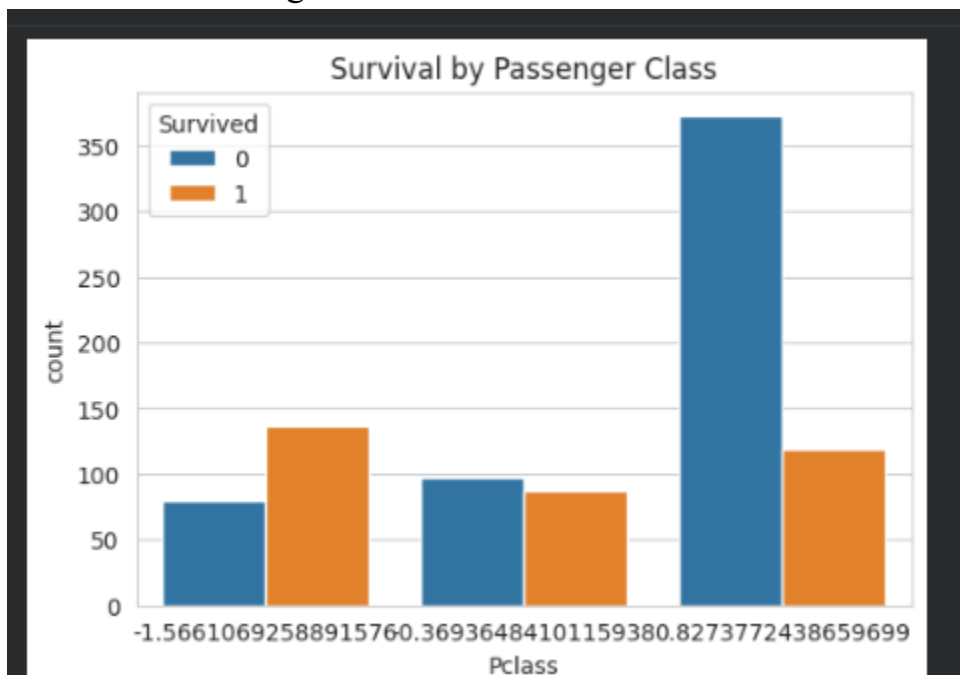
- Gender Countplot



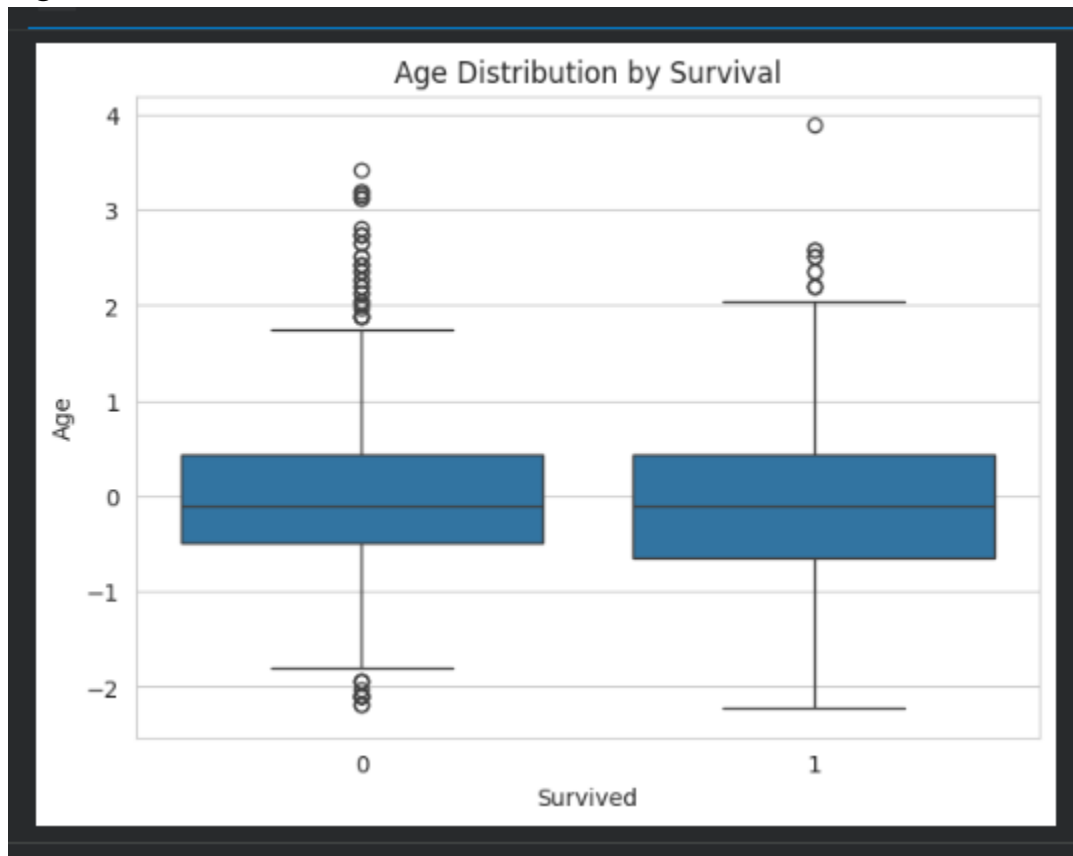
- Survival vs Gender



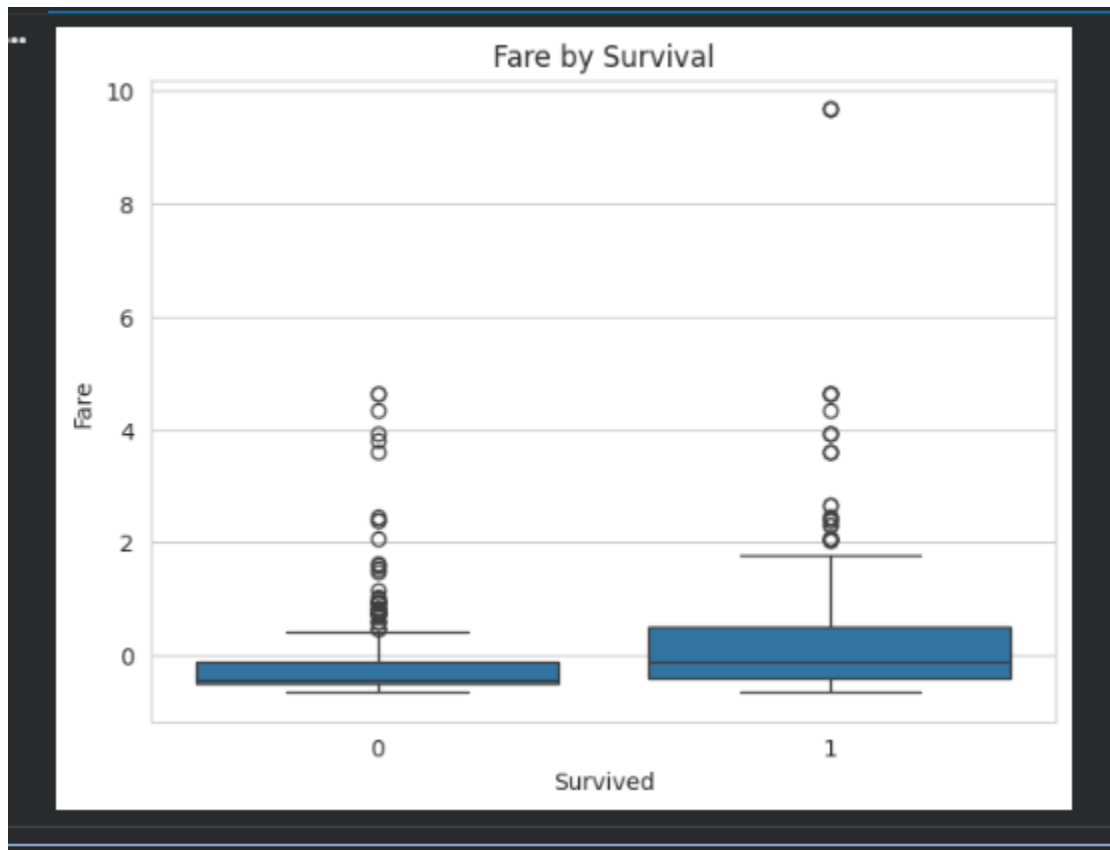
- Survival vs Passenger Class



- Age vs Survival



- Fare vs Survival



Assignment Questions

Which feature has the highest variation?

Fare shows the highest variation among numerical features.

Are there any skewed distributions?

Yes. Fare exhibits a positively skewed distribution.

Which category occurs most frequently?

Passenger Class 3 contains the highest number of passengers.

Which variables are most related to the target variable?

Sex, Passenger Class, and Fare show the strongest relationships with passenger survival.

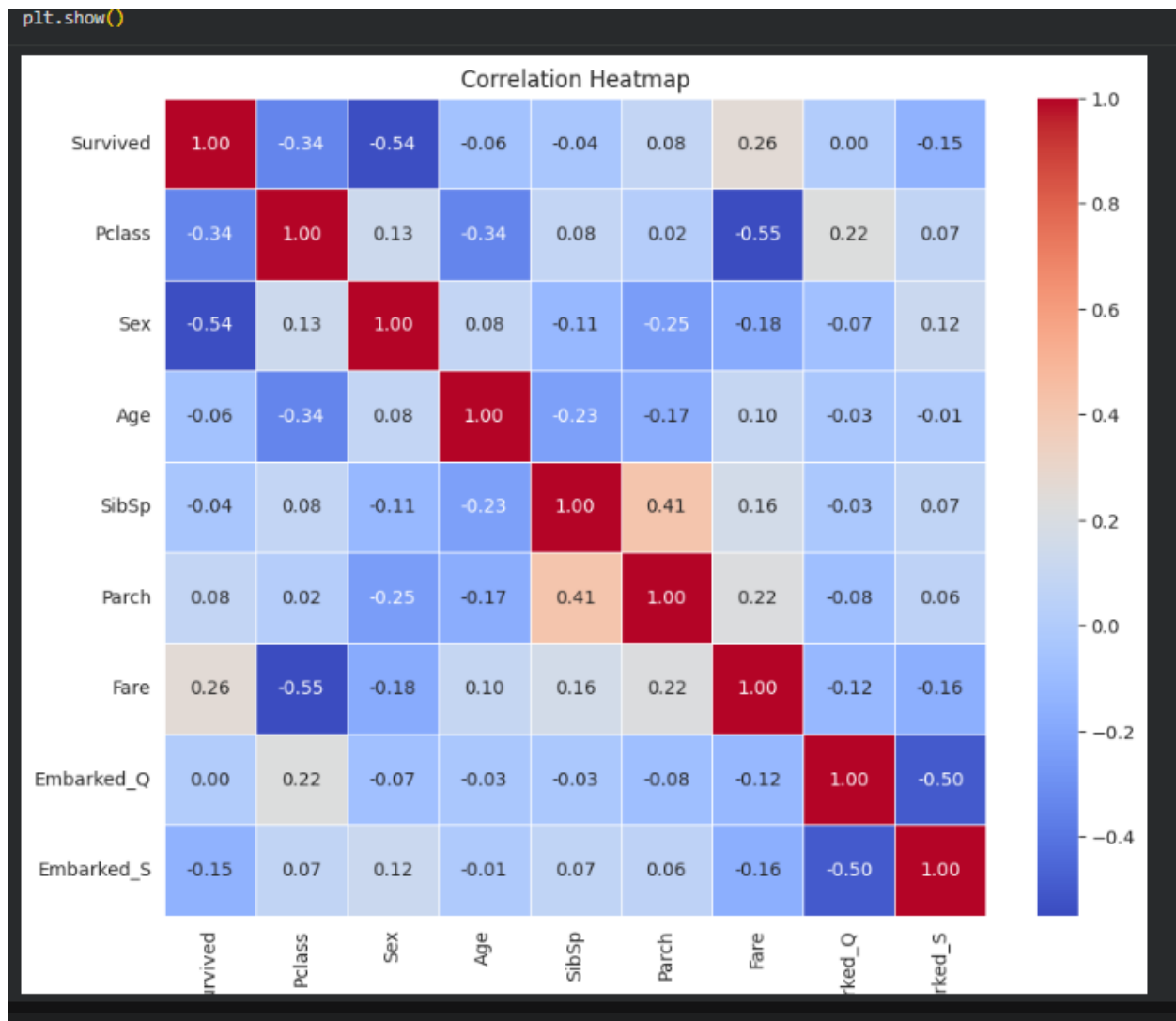
What trends were observed?

- Female passengers survived more frequently than males.
- First-class passengers had a higher survival rate.
- Passengers with higher ticket fares were more likely to survive.
- Most passengers traveled in third class.
- Age alone showed a weaker relationship with survival.

6. Correlation Analysis

A correlation matrix and heatmap were generated to examine relationships between numerical variables. The analysis showed that Fare has a positive relationship with survival, while Passenger Class has a negative relationship with Fare because first-class passengers generally paid higher ticket prices.

No extremely high correlations were observed among the remaining features, indicating that most variables contribute unique information and are suitable for machine learning models.



7. Machine Learning Readiness Summary

The Titanic dataset has been successfully prepared for machine learning model development.

The following preprocessing steps were completed:

- Missing values in Age were replaced using the median.
- Missing values in Embarked were replaced using the mode.
- The Cabin column was removed due to excessive missing values.
- No duplicate records were found.

- Outliers were analyzed and retained because they represented genuine observations.
- Label Encoding was applied to the Sex column.
- One-Hot Encoding was applied to the Embarked column.
- StandardScaler was used to normalize numerical features.

After preprocessing, the dataset contained no missing values or duplicate records and was ready for future machine learning model training.

8. Key Insights

1. Female passengers had a significantly higher survival rate than male passengers.
2. Passengers traveling in first class were more likely to survive than those in lower classes.
3. Fare displayed the highest variation and was positively associated with survival.
4. Most passengers traveled alone or with a small family, as reflected by the SibSp and Parch features.
5. After preprocessing, the dataset became clean, consistent, and suitable for machine learning model development.

9. Conclusion

This project successfully demonstrated the complete data preprocessing and exploratory data analysis workflow for the Titanic dataset. Missing values, duplicate records, categorical variables, outliers, and feature scaling were handled using appropriate techniques based on the characteristics of the dataset.

Exploratory data analysis provided valuable insights into passenger survival patterns, highlighting the influence of gender, passenger class, and ticket fare. The final cleaned dataset is fully prepared for machine learning model development in subsequent stages of the internship.